



How Artificial Intelligence Technology Affects Productivity and Employment: Firm-level Evidence from Taiwan

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ABSTRACT

The effects of the rapid development of artificial intelligence (AI), a general-purpose technology, on firm performance is an emerging and crucial issue. This study examines the impact of AI technology on firms' productivity and employee profiles. We use the keyword-matching method to parse the text of Taiwan patent grants, and obtain matched firm-level data on AI innovations in Taiwan's electronics industry for the 2002–2018 period. Empirical estimations indicate that AI technology is positively associated with productivity and employment. Meanwhile, non-AI patents also generate pro-productivity and pro-employment effects with a magnitude similar to that of AI technology. Inventing AI technologies crucially alters firms' workforce compositions, which reduce the share of labor force with educational qualifications of college level and below. Robustness checks reaffirm these findings.

1. Introduction

Artificial intelligence (AI) has witnessed rapid advancement in the past decade alongside related technologies such as machine learning, robotics, and neural networks. Since AI has potential as a new general-purpose technology (GPT) (Trajtenberg, 2018), breakthroughs in AI-related technologies (hereafter AI technology) have revolutionized the development of new manufacturing technologies and products that can be applied to a wide range of fields, such as the Internet of Things, dark factories, autonomous vehicles, remote medical monitoring/diagnosis systems, and others. The AI revolution has induced manufacturing sectors to head toward intelligent manufacturing, which is undergoing continuous development.¹ Undoubtedly, the pace and scope of groundbreaking development in AI technology can substantially change firms' production methods, thereby affecting firm productivity. Meanwhile, there are wide concerns and increasing debates on its potential to cause job loss and wage stagnation for most workers (Brynjolfsson and McAfee, 2015). Therefore, examining the influence of AI technology on firms' productivity and employment is crucial and topical.

Numerous studies have examined the productivity and employment effects of innovations on firms in the past few decades. A technological change is expected to enhance productivity which is a measure of a

technology's economic contribution. However, firms may not obtain productivity gains through new technologies, even for a GPT. The "productivity paradox" claimed in Brynjolfsson (1993) indicated that while firms had an appetite for investing in information and communication technology (ICT), which was a GPT in the 1990s, it caused insignificant productivity gains. Possible causes include learning and adjustment, ICT management, and redistribution and dissipation of profits.² As reviewed by Mohnen and Hall (2013), various innovation types—product, process, organizational, and marketing innovations—contribute significantly to firm productivity, whereas some studies find an insignificant or negative relationship.

The influence of innovation on employment is controversial. Product innovation can create new jobs (or at least does not cause job destruction), but it could possibly lead to a polarization of jobs, whereas process innovations (generally labor-saving) are mostly job-destroying. (Dosi and Mohnen, 2019; Staccoli and Virgillito, 2021). The job-creation effect of innovation is particularly relevant to high-tech firms (Van Roy et al., 2018). Distinguishing embodied from disembodied technological changes, their employment effects also differ (Dosi et al., 2021). A GPT, such as computerization, may put approximately half of all occupations at risk (Frey and Osborne, 2017), but workers' job duration depends on their skills and the technology-skill complementarities (Castro Silva and

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¹ The market size for AI worldwide is expected to grow annually from around US\$ 5 billion in 2015 to over US\$ 125 billion in 2025, as forecasted by Statista. <https://www.statista.com/statistics/941835/artificial-intelligence-market-size-revenue-comparisons/>

² Another pure research issue is mismeasurement of outputs and inputs.

Lima, 2017).³

Being an emerging and influential GPT, does AI technology promote firm productivity or does it also suffer from the “modern productivity paradox” (Brynjolfsson et al., 2019)? AI enables firms to find productive new combinations of existing technologies (Agrawal et al., 2019a), thus improving firm productivity. AI can be the command center of a manufacturing process management system to improve the efficiency of input usage, and to analyze and adjust equipment performance from data. Machine learning is suited for augmenting or automating tasks that involve at least some prediction aspects (Agrawal et al., 2017); that is, the automation production that was “programmed” by engineers can be replaced by machine learning or deep learning designed to learn the production patterns automatically from data (Brynjolfsson and Mitchell, 2017). Brynjolfsson et al. (2018) also indicated that AI machine vision systems have a lower error rate than humans for the same dataset. Therefore, the manufacturing instruments, i.e. robots, can perform a variety of hand-eye coordination tasks rather than routine machine actions (Levine et al., 2016). Thus, it helps to improve productivity.

Alternatively, AI technology may not generate productivity gains immediately upon commissioning (Jovanovic and Rousseau, 2005). The potential time lag in the productivity effect of AI technology may be attributed to adjustment costs, complementary innovations, and organizational changes (Brynjolfsson et al., 2019). In practice, the adjustment and combination of AI and existing technologies may not be successful. AI technology is not a panacea that can be applied to all production processes and firms. For instance, some sudden occurrences of production process might be managed and judged better by experienced workers; the magnitude and level of AI technology introduced into the production process may not fit well with the existing manufacturing workforce and technologies. This is because the effective use of AI technology relies heavily on big data that must adapt to firm-specific human capital and manufacturing processes. Workers often have no incentives to learn them, being afraid of losing their jobs.

There are wide concerns and increasing debates on its potential to cause job loss and wage stagnation for most workers (Brynjolfsson and McAfee, 2015). The introduction of AI technology casts the fear of an era of mass joblessness. Continuing progress on the predictive power of AI, the so-called deep learning technique, which simulates the human brain by learning from big data, empowers AI to replace humans in decision-making across many professional areas. The scope of replacement includes not only manual tasks but also intangible inputs (Jones and Romer, 2010) and organizational processes (Jarrahi, 2018).

AI technology can generate 3D virtual paths and programmable intelligence and enable machines to think intelligently in coordination with the simulation of many examples or data (Goldfarb and Trefler, 2018). Thus, AI may perform better than humans using the deep learning model to analyze previous examples/data and simulate various situations based on big data. For example, the IBM Deep Blue first won against a world champion in the 1996 chess match by evaluating the function from past board game records. A significant advance is Google AlphaGo: this supercomputer completed self-learning through an artificial neural network (a deep machine learning method) to improve chess-playing capability and beat the top-ranked chess player in the world in 2017. AI has also penetrated many organizational processes (Jarrahi, 2018). Through the deep learning algorithm system, AI has developed strength in dealing with organizational decision-making processes that are uncertain, complex, and equivocal. Thus, AI technology can replace not only manual and routine tasks conventionally performed by humans, but also tasks that are based on human judgment and decisions in manufacturing and management processes.

Brynjolfsson and McAfee (2015) and Agrawal et al. (2019b) thought

that there is no obvious answer to the question of the AI-employment nexus. While AI technology may destroy some jobs, it may increase jobs at the low and high ends from an aggregate perspective. From the firm perspective, if AI technology supplements and amplifies tasks performed by workers, both production efficiency enhancement and product extension effects might generate more jobs. For example, although robots are flexible, versatile, and autonomous machines, the adoption of robots relies on their comparative advantage in specific tasks. Robots did not significantly reduce total employment, although they did reduce the employment share of low-skilled workers (Graetz and Michaels, 2018). Thus, the employment effects of AI technology are probably task- and skilled-based, altering the allocation of tasks to labor and capital of production (Acemoglu and Restrepo, 2019).

Although there are widespread concerns and debates on how AI technology affects firm productivity and employment, firm-level evidence is scarce. Therefore, more firm-level analyses on AI invention and its effects are required (Raj and Seamans, 2019). To fill this gap, this study uses a firm-level panel dataset of Taiwan's electronics firms to examine the impact of AI technology on firm productivity, employment, and workforce composition. This study contributes to innovation literature in the following ways. First, it provides one of the first pieces of evidence on how AI technology affects firm productivity and employment. Specifically, we examine the potential impact of AI technology introduction on workforce composition. Quantifying the impacts of AI technology can have policy implications for governments to ameliorate its potential impacts on the labor market. Second, we employ the keyword-matching approach to identify patents that describe AI technology. The U.S. Patent and Trademark Office (USPTO) has created several hundred patent classes, including AI-related patent classes, such as 706 (data processing: artificial intelligence) and 901 (robots). However, using only patent counts from limited AI-related classes may undercount the true scope of AI-related inventions (Alderucci et al., 2020). This keyword-matching approach identifies AI technology patents more precisely and is applicable to patent offices that adopt only the International Patent Classification (IPC) system, which is more adequate and has been adopted in some prior work in this area (e.g., Keisner et al., 2016; De Prato et al., 2019; Van Roy et al., 2019). Third, we compare the productivity and employment effects between AI and non-AI patents. Considering the fast technological change, the advance of non-AI technology can also have considerable impacts on firms alone, as examined in existing studies. Ignoring the impact of non-AI patents may overestimate the economic impact of AI technology.

The remainder of this paper proceeds as follows. Section 2 provides an overview of AI technology patents and reviews related literature. Section 3 describes the data and some stylized facts. Section 4 presents the empirical results of the link between AI technology and productivity in Taiwan's electronics firms. Section 5 presents the influences of AI technology on employment and workforce composition. Concluding remarks are summarized in the final section.

2. Overview of AI technology patents and literature review

2.1. AI technology and AI-related patents

What is AI technology? There are numerous and various definitions. According to seminar research in Nilsson (1980), “*The fundamental AI ideas underlie applications such as natural language processing, automatic programming, robotics, machine vision, automatic theorem proving, and intelligent data retrieval, rather than focusing on the subject matter of the applications.*” It is a loose term used to describe technologies with human-like intelligence (Raj and Seamans, 2019). The hot AI technology includes virtual agents, machine learning platforms, AI-optimized hardware, decision management, deep learning platforms, robotic process automation, text analytics and natural language processing (NLP), and others (Press, 2017). In other words, the AI analytical framework can be categorized into three layers adopted from the World Intellectual

³ The impacts of ICT on firm productivity and employment have been widely examined in literature, but results are inclusive. Please see comprehensive reviews in Cardona et al. (2014) and Calvino and Virgillito (2018).

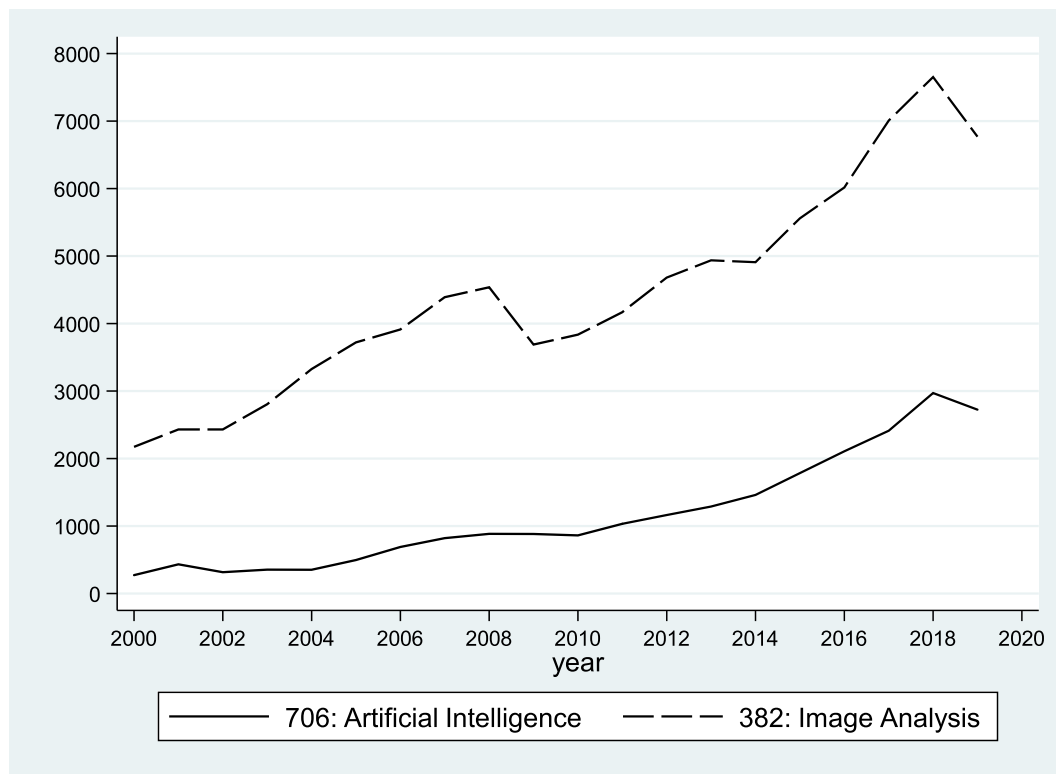


Fig. 1. AI-technology Patents by Filing Year. Source: Patent Examination Data System, USPTO.

Property Office (WIPO), namely methodology, function, and applications.

We start by noting the trend of two classes of AI technology patents filed in the USPTO. US patent class (USPC) 706: Artificial Intelligence, is directly relevant to AI technology; USPC 382: Image Analysis, which is useful in conducting trend analyses. Fig. 1 presents the trend of AI technology patents; USPC 706 and USPC 382.

In terms of class 706, this graph illustrates a dramatic increase in the number of AI technology patents, particularly since the late 2000s onward. There were 273 filing patents in 2000 and this number increased to 2725 in 2019. The patent count for Image Analysis technology also exhibits an increasing trend from 2173 in 2000 to 6770 in 2019. However, the number decreased considerably in the period 2009–2011. WIPO (2019) also indicated that computer vision (which includes image recognition), is a crucial component in patents of AI functional applications.

Machine learning (e.g., deep learning and neural networks) is not coded in the USPC, but this technique can be applied to various technology domains. According to WIPO (2019), machine learning dominated the patent field for AI techniques, both in terms of absolute numbers and year-on-year growth rates in 2013–2016. The plurality (40%) of the AI patents relate to machine learning; patent filings for machine learning experience a 28% growth per year on average. Therefore, the limited AI-related classes in the USPTO may cause underestimation of AI inventions, as argued by Alderucci et al. (2020).

Which countries are the leaders in AI technology inventions? Is Taiwan functional as a typical case study for examining the impact of AI technology on firms? We use geographic data for the assignees or inventors of patents granted by the USPTO to briefly compare the performance of AI technology across countries. Fig. 2 depicts the geographic distribution of inventors patented under USPC 706 (AI) and 382 (image analysis) patents from 2000 to 2019. The US inventors account for 58.35% of AI patents (class 706) issued by the USPTO. Although there is home bias in that US inventors tend to apply for patents within the US first, the US seems to dominate AI technology

inventions in the world. There is a widespread concern about China's challenge to become a superpower in AI technology. The graph (black bar) demonstrates that of the total number AI patents, 8.29% were issued to inventors in Japan, 4.56% to inventors in Germany, and 3.44% to inventors in the UK. China accounted for only 2.40% of AI patents remaining far behind the US, Japan, and Germany. Taiwan ranked tenth and accounted for only 1.07% of the patents; however, this performance is outstanding if the number of patents is deflated by population.

The top two countries in the category of image analysis (class 382) patents in the USPTO are the US and Japan, which accounted for 35.75% and 29.62%, respectively. This shows that after considering home bias in patenting, Japan has a strong technological capability in AI technology. Other countries are far behind the leading group, and the second-tier group comprises Korea (4.56%), Germany (4.18%), and Taiwan (3.11%).

2.2. Literature review

Does AI technology bring a new era of productivity growth or suffer from the modern productivity paradox? Does it really destroy jobs, as argued by skeptics? These topical issues have attracted an emerging line of literature, as most studies focus on specific AI-related technologies or products. The cross-country analysis by Graetz and Michaels (2018) finds that robot adoption within industries raises labor productivity and total factor productivity (TFP). Ballestar et al. (2020) link robots to labor productivity in Spanish small and medium manufacturing enterprises (SMEs). Adopting a dichotomous variable to examine whether using robots can enhance performance, they found associations between robotic devices and higher productivity and employment rates. Digital adoption is also associated with productivity gains at the firm level in EU countries, with a stronger effect on manufacturing and routine-intensive activities (Gal et al., 2019).

Regarding the employment effect of AI technology, Acemoglu and Restrepo (2020) find a large and robust negative effect of robots on employment in the US: one more robot per thousand workers reduces

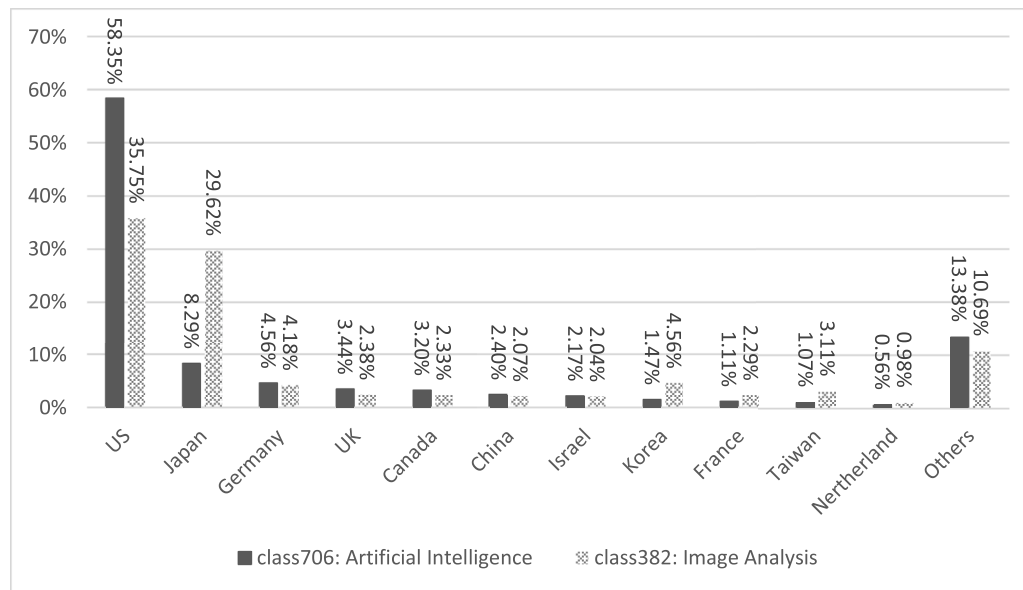


Fig. 2. AI-technology by Country, 2000–2019. Source: calculated by author from the USPTO.

the employment to population ratio by about 0.18–0.34 percentage points. However, [Graetz and Michaels \(2018\)](#) argue that robots overall have no significant association with employment, but they reduce the use of low-skilled workers. [Felten et al. \(2019\)](#) investigate the link between AI and labor by creating a new measure called AI occupational impact. Using information about occupational tasks in American workplaces, the estimation results indicate that AI does not affect employment. Leveraging a global survey of 3000 firms across 10 countries, [Bughin \(2020\)](#) constructs AI occupation-level impact scores to examine how AI tools affect demand for labor and the related mix of skills. The estimation results show that AI investment growth is likely to increase employment, rather than reduce employment levels.⁴

As mentioned previously, a growing number of studies use the number of patents as an indicator of AI technology (e.g., [Cockburn et al., 2019](#); [Webb et al., 2018](#)), but firm-level evidence on the relationship between AI technology and productivity or employment is limited. [Alderucci et al. \(2020\)](#) provide the first firm-level evidence with a holistic measure of AI technology to quantify the impact of AI on productivity and employment. Identifying AI-related patent grants by the USPTO and matching them with the US Census microdata, results obtained from the event study obtain two findings. First, AI-related innovations appear to raise firms' labor productivity in terms of output per worker. Second, AI-related innovations are positively associated with employment growth, as they result in 25% faster growth. [Damioli et al. \(2021\)](#) examine the relationship between AI and labor productivity using a worldwide sample of 5,257 companies that have filed at least one AI patent. The results show that AI patent applications have an extra-positive effect on labor productivity, particularly for SMEs and services industries.

3. Data source and overview

3.1. Identifying AI patents

The first challenge in studying the impact of AI technology on firm performance is the measurement of AI technology. Patent information is a feasible source of data because firms have a strong incentive to patent

their AI-related technology to prevent imitation, block potential competitors, and appropriate commercial profits.

A natural approach is to extract AI-related patents from the Taiwan Patent Office (TPO). However, unlike the USPTO, which maintains a detailed taxonomy containing patent classes of AI-related technology,⁵ patents granted by TPO are classified according to IPC; that is, patents are classified according to the areas of technology to which they pertain, and there are no specific classes for different types of AI technology patents. Since AI innovations are filed across a wide range of technologies, this study adopts the keyword-matching approach used by other studies in this area ([Keisner et al., 2016](#); [De Prato et al., 2019](#); [Van Roy et al., 2019](#)). This method enables us to identify patents that describe the use of particular algorithms, techniques, or areas of AI-related technology.

The choice of keywords was based on WIPO's three-layer AI analytical framework (methodology, function, and applications), US patent classes for AI technology, and consultations with scientists and engineers. We searched for patents containing a number of specific keywords (see Appendix Table A1) and included them as AI innovations. This keyword-matching method uses stringent definitions of AI relatedness and can achieve high precision in identifying AI patents. However, it may not detect all AI-related patents because while some inventions are probably AI technologies, they do not contain easily identifiable keywords in patent content ([Alderucci et al., 2020](#)).⁶ As there is no perfect measure for AI technology in practice, we might have underestimated the number of AI technology patents, implying that estimates of the impacts of AI technology could be up-biased. Appendix Table A2 summarizes the IPC classifications assigned to AI innovations in Taiwan.

Based on the above IPC codes for AI technologies, [Fig. 3](#) depicts the AI invention number and its ratio to total inventions in Taiwan in 2000–2019. Only 243 AI inventions were granted patents in 2000, which increased to 2,991 in 2009. After a slight decline in 2010–2011, it increased from 3138 in 2012 to 4973 in 2019. The corresponding ratio of AI patents to total inventions increased from 1.55% in 2000 to 6.00% in 2019.

⁵ [Cockburn et al. \(2019\)](#) and [Webb et al. \(2018\)](#) identify AI-related inventions from USPC.

⁶ [Alderucci et al. \(2020\)](#) use a combination of natural language processing and machine learning algorithms to identify AI-related innovations by parsing the text of US patent grants.

⁴ Existing evidence regarding the effects of robots and/or AI on aggregate labor and productivity provides a mix view. See [Raj and Seamans \(2019\)](#) for a review on studies using country, region, or industry data.

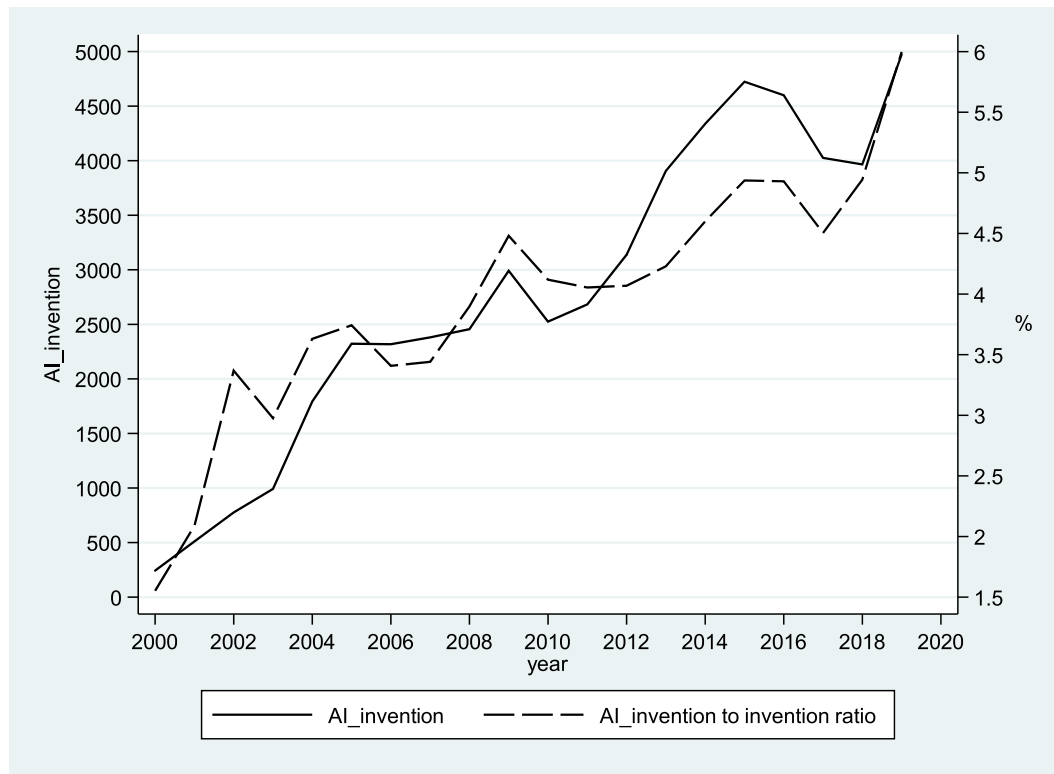


Fig. 3. AI Inventions in Taiwan, 2000–2019.

3.2. Firm-level dataset

The microdata analyzed in this study are a firm-level longitudinal dataset of Taiwanese electronics firms, which are listed on the Taiwan Stock Exchange (TSE) or the Over-the-Counter (OTC) market for the 2002–2018 period. The rich microdata constructed and maintained by the Taiwan Economic Journal (TEJ) provide information in three areas: (1) basic information such as company name, establishment date, and number of employees; (2) information from financial statements; and (3) production information such as sales, output, R&D expenditure, and exports. Firms listed on the TSE and OTC are mostly large-and medium-sized enterprises (LMEs) due to the initial public offer rule. Thus, this dataset is suitable for investigating the impact of AI technology on firm performance in Taiwan because of two reasons.

First, as displayed in Appendix Table A2, matched patents are classified into section G: physics of IPC codes. The broad description of the contents of section G suggests that most of them should be filed by electronics firms. The electronics industry is not only the key industry in Taiwan's manufacturing sector in terms of output and R&D spending,⁷ but also plays a key role in the global electronics production chain, particularly for products such as semiconductors, medical engineering applications, and informational and communication technologies. Second, firms with less than 200 employees, the so-called SMEs, account for an overwhelmingly large share of the manufacturing sector in terms of the number of firms, which reached more than 96% in the past decades. However, small firms (less than 100 employees) demonstrate a low propensity to undertake R&D (4.99% in 2006), whereas the corresponding propensity for R&D in large firms was 77.46% (Yang and Tsou, 2016). Therefore, LMEs are more suitable for investigating the impact of AI technology on firm performance.

3.3. Overview of AI-inventing firms

We link the company names to the TPO patent databank, to obtain individual firms' patent information, including AI technology patents. Table 1 provides an overview of AI-innovating and non-AI-innovating firms. We categorize electronics firms without patents as typical electronics firms, firms with non-AI patents as innovating firms, and firms with at least one AI patent as AI-innovating firms. This classification helps to determine the differences between AI-innovating firms and others.

Innovating firms are the majority group in the electronics sector, accounting for 51.90% of firm observations, followed by typical firms (35.40%). Firms with AI patents remain a minority, accounting for 12.70% of the firm observations. Thus, patenting activity prevails in the electronics sector, reaching 64.60%. This is because patents not only promote productivity but also block potential competitors in this technologically dynamic industry. Although AI technologies concentrate on a small number of firms, AI-innovating firms are more productive, larger, pay higher salaries, and devote more R&D investment than their typical and innovating counterparts. Typical firms have the smallest firm size, pay the lowest payroll per employee, and have the least R&D expenditures. Moreover, firms with AI patents are more aggressive in internationalization activities in terms of export intensity and the establishment of foreign affiliates relative to their counterparts. Notably, the t tests in Column 5 indicate that AI-innovating firms have higher productivity and have significant differences in most firm characteristics relative to innovating firms (typical firms).

4. AI technology and productivity

4.1. Baseline model

To assess the impact of AI innovations on firm productivity, the empirical model is specified as follows:

⁷ For instance, the electronics industry output and R&D expenditure accounted for 38.78% and 80.75% in the manufacturing sector in 2015.

Table 1
Summary Statistics by Firm Type, 2002–2018

Innovation Types	(1) Entire Sample	(2) Typical firms	(3) Innovating firms	(4) AI-innovating firms	(5) t-test (4) vs. (3)
Productivity	11.407	11.066	11.375	12.484	27.319***
Mean Employment	728.843	307.781	619.500	2,348.838	24.131***
Mean Age	21.327	21.700	21.211	20.765	1.377
Mean wage	729.293	659.366	716.961	977.408	22.808***
Export propensity	71.808	68.319	72.355	79.216	8.937***
Foreign affiliates	1.816	1.293	1.563	4.310	15.407***
R&D expenditure	377.654	52.599	177.160	2,096.819	24.446***
Observations	10,651	3,770	5,528	1,353	

Note: Productivity is calculated using the semi-parametric estimates developed in [Levinsohn and Petrin \(2003\)](#). The numbers displayed are based on employment and age. The observation numbers of other variables are smaller slightly due to the missing problem in some firm observations.

*** $p < 0.01$.

$$Productivity_{it} = \alpha_0 + \alpha_1 AI_{it} + \gamma X_{it} + u_t + v_i + \varepsilon_{it} \quad (1)$$

where the outcome variable *Productivity* is the TFP of firm *i* in year *t*. It is measured using the semi-parametric method developed by [Levinsohn and Petrin \(2003\)](#).

As the calculated TFP is a logarithmic form, explanatory variables are mostly measured by the forms of logarithm or ratio, enabling us to interpret the estimated coefficients as productivity elasticities of individual factors. AI is a key covariate of concern that denotes AI innovations. We use two measures: treatment measure (0/1) and the number of AI patents. *X* is a vector of firm characteristics that influence firm productivity. Notably, we also consider other patents (non-AI) which are key drivers of firm productivity ([Mohnen and Hall, 2013](#)). Other firm characteristics include firm size (the logarithm of number of employees), firm age, export ratio (export to sales ratio), and number of foreign affiliates. Term *u* is the year fixed effect, denoting the macro-economic shocks affecting firm performance. Finally, *v* and *ε* denote the unobserved firm heterogeneity and error terms, respectively. [Table 2](#) presents the definitions and summary statistics of the variables.

[Table 3](#) reports estimation results of various specifications using the fixed effect of panel data model. As shown in Column 1, the presence of AI patents has a significantly positive influence on productivity. Firms with AI patents are associated with a 13.07% higher productivity than other firms, without controlling for non-AI patents and firm characteristics. As AI and non-AI patents contribute to promoting productivity, we include both AI and other patent numbers in Column 2. The estimates show that both AI patent and non-AI patents positively and significantly impact productivity with elasticities of 7.68% and 8.77%, respectively. Although the non-AI patent variable is associated with a larger elasticity than that of the AI patent, their difference in economic magnitude is insignificant.

The main results are shown in Columns 3 and 4, which include firm characteristics. The variables of AI and non-AI patents are still associated with a significantly positive coefficient. According to the coefficient values in Column 3, a 10% increase in AI patents induces a 0.589% increase in productivity, which is larger than that induced by non-AI patents (0.444%). Column 4 also shows that electronic firms with AI patents and those with non-AI patents are respectively associated with 7.85% and 5.04% higher TFP than non-innovating firms, while the difference in productivity effects of AI and non-AI patents are statistically insignificant. This result highlights the importance of AI technology development in promoting productivity, which echoes the conceptual prediction in [Agrawal et al. \(2019c\)](#) and is consistent with the findings of [Alderucci et al. \(2020\)](#) and [Damioli et al. \(2021\)](#). AI technology can be a core production method either independently or by complementing existing technologies, thereby promoting productivity. Meanwhile, other patents can lower production costs and/or increase product quality, and their productivity effect is also considerable.

The effects of firm characteristics are significant: larger and older electronics firms exhibit higher productivity. International trade and

Table 2
Variable Definitions and Basic Statistics

Variable	Definition	Mean (S. D)
Productivity	Total factor productivity, measured by method of Levinsohn and Petrin (2003)	11.407 (1.384)
AI treatment	AI technology dummy: 1 = with positive AI patents, 0 = without AI patent	0.127 (0.333)
AI patent	Number of AI patents granted	1.180 (10.434)
Other patent treatment	Other patent dummy: 1 = with positive patents rather than AI patents, 0 = without other patent	0.642 (0.479)
Other patent	Number of other patents	21.071 (133.556)
Firm size	Measured by the number of employees	728.843 (2,023.92)
Firm age	Data year minus the established year	21.327 (10.658)
Export intensity	The export to sale ratio (%)	71.808 (27.047)
FDI	Number of foreign affiliates	1.816 (4.955)
Firm sale	Sales (NT\$ billion)	15.701 (113.935)
R&D expenditure	R&D expenditure (NT\$ million)	377.654 (2,182.59)
Low-skilled worker	The ratio of employees with less than college education (%)	50.077 (18.317)
High-skilled worker	The ratio of graduate level-educated employees (%)	11.526 (13.468)

Note: The basic statistics of firm characteristics are calculated using data for 2002–2018.

investment also affect firm productivity. Export intensity positively impacts productivity because fierce competition in foreign markets facilitates exporters to promote productivity. We find that outward FDI, in terms of affiliate numbers, is positively associated with productivity, which is consistent with the findings of [Huang et al. \(2013\)](#) for Taiwan's electronics industry.

4.2. Measurement issues

To ensure that AI technology facilitates productivity enhancement in inventing firms, we address two potential issues. One is the measurement issue, that is, the inclusion of the measures of productivity, novelty of patents, and the inclusion of outliers. The second issue is the potential endogeneity problem, which stems from the potential reverse causality from productivity to innovation.

We address the measurement issue in three dimensions. First, if AI raises TFP, it may also increase labor productivity. The explained variable TFP is thus replaced by labor productivity, which is measured as the logarithm of value added per employee. Second, patents are categorized

Table 3
AI Technology and Productivity

	(1)	(2)	(3)	(4)
AI treatment (1/0)	0.1307*** (0.0329)			0.0785** (0.0324)
AI number (ln(AI+1))		0.0768*** (0.0241)	0.0589** (0.0237)	
Other patent treatment (1/0)				0.0504*** (0.0226)
Other patent number (ln(patent+1))		0.0877*** (0.0108)	0.0444*** (0.0110)	
Firm size (ln(employee))			0.2217*** (0.0183)	0.2379*** (0.0179)
Firm Age			0.0413*** (0.0030)	0.0420*** (0.0030)
Export intensity			0.0068*** (0.0006)	0.0068*** (0.0006)
Number of foreign affiliates			0.3611*** (0.0396)	0.3691*** (0.0396)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R-squared	0.174	0.181	0.212	0.211
Observations	10,651	10,651	10,582	10,582

Note: Figures in parentheses are robust standard errors. *** $p < 0.01$, ** $p < 0.05$.

into three types: invention, utility, and new design. The invention patent, which is associated with new technical solutions for products and processes, is considered to have greater technological novelty and granted a longer legal protection period of 20 years. The primary variables of AI and other patents are re-measured by including only invention patents. Third, we exclude two firms with extremely large numbers of AI patents to examine the robustness of the AI–productivity nexus.⁸ Table 4 presents the estimation results.

AI patents overall have a significantly positive effect on productivity. When we replace TFP with labor productivity, Column 1 shows that the economic magnitude of the effects of AI on labor productivity is smaller (0.0374) than that of TFP (0.0589), as shown in Table 3. This result implies that AI technology has a greater effect on capital productivity. In Column 2, after controlling for other patents, the estimated coefficient of the AI dummy is 0.0796. Although the estimated magnitude is slightly smaller than that for US firms (0.0890), as reported by Alderucci et al. (2020), their estimations did not control for other patents.

Using higher novelty of invention patents as innovation indicators, Column 3 shows that the economic magnitudes of each AI patent and other patents on TFP become smaller or larger than the estimates in Table 3. However, based on the results in Column 4, we find that both impacts of the presence of AI patents on TFP become larger. Finally, Columns 5 and 6 demonstrate that the estimation results are less sensitive to the inclusion or exclusion of outliers because the estimated coefficients of AI and other patents remain similar.

4.3. Endogeneity problem

There is also an endogeneity issue that not only firms with certain characteristics may self-select to develop AI technologies, but there is also a reverse causality from AI technology to firm characteristics. Moreover, other control variables can also be endogenous, which can cause inconsistency in estimation results. As displayed in Table 1, AI-inventing firms demonstrate considerable differences in firm characteristics and perform better in productivity compared to other firms.

⁸ The two firms excluded are Hon Hai technology group (TSE code: 2317, the parent firm of Foxconn) and Invent Corporation (TSE code: 2356). Their mean AI patents are, respectively, 172.29 and 105.32, which are three times larger than the standard deviation of AI inventing firms (mean = 9.292, standard deviation = 27.968). Estimation results are similar when we only exclude Hon Hai technology group.

This endogeneity problem and confounding factors may prevent an effective direct comparison between AI-inventing firms and non-AI-inventing firms. Conventionally, the propensity score matching (PSM) approach to conduct an adequate “like-for-like” comparison can mitigate the selection bias that firms are not randomly distributed, but it may not well tackle the endogeneity problem.⁹ Considering that productivity is probably path-dependent, we employ the generalized method of moments (GMM) for the dynamic panel data model to ensure the robustness of the results.

The specification of the panel dynamic model includes the lagged dependent variable as follows:

$$Productivity_{it} = \alpha_0 + \beta_1 Productivity_{i,t-1} + \alpha_1 AI_{it} + \gamma X_{it} + u_i + v_i + \varepsilon_{it} \quad (2)$$

Difference GMM estimation (GMM-DIF), which was developed by Arellano and Bond (1991), and system GMM (GMM-SYS) estimation, introduced by Arellano and Bover (1995) and Blundell and Bond (1998), are feasible methods for tackling endogeneity, although each technique has its own disadvantages. In a dynamic panel model, GMM-DIF may suffer from small-sample bias if the time span is too short (Alonso-Borrego and Arellano, 1999). Although GMM-SYS estimators may be more efficient than GMM-DIF estimators, when a series features strong autocorrelation and cross-sectional variability is dominant, the proliferation of instruments in GMM-SYS may lead to finite-sample bias owing to the overfitting of endogenous variables (Roodman, 2009a). Therefore, this study employs both methods to conduct robustness checks. Table 5 presents the estimation results.

Since GMM-SYS and GMM-DIF may both easily generate invalid estimates (Roodman, 2009a), the study first uses the instrumental variable of a dynamic panel data model, a special GMM case. The results are provided in Column 1. The variable of 1-year lagged productivity is associated with a significantly positive coefficient of great magnitude, indicating that productivity exhibits path-dependence. We find that AI patents continues to have a significantly positive influence on productivity, whereas other patent is associated with an insignificant coefficient. Possible endogeneity is considered only for AI technology but not for the other covariates.

Columns 2 and 3 employ GMM-DIF estimators. The results for Hansen-Sargan testing, which are not significant, thereby indicating the adequacy of this approach, appear at the bottom row. The values reported for AR(1), AR(2), and AR(3) are the p -values for the first-, second-, and third-order autocorrelated disturbances in the first-differenced equation. Owing to an elevated level of second-order autocorrelation, the estimates in Column 3 are obtained by including the lagged one- and two-period dependent variables as covariates.

Considering the dynamic nature of productivity and the endogeneity issue, we find that the estimates for AI technology are similar: AI patents continue to be associated with significantly positive coefficients in Columns 2 and 3. The estimates in Column 3 reveal that the productivity increases by 0.401% if the number of AI patents increases by 10%, which is less than the increase reported in Column 3 of Table 3 (0.589%). However, this productivity effect is greater than that induced by non-AI patents (0.262%).

The GMM-SYS estimators are more efficient; therefore, we present them in Columns 4 and 5. The estimated coefficient of AI patents in Column 4 is positive but statistically significant. After dealing with the second-order autocorrelation problem, the estimates in Column 5

⁹ We also adopt the PSM approach to tackle the sample selection problems and obtain similar findings that AI patents have a positive association with productivity in the current and future two years. Please see the online Appendix Table A3.

Table 4
AI Technology and Productivity—Measurement Issues

Dep. variable	(1)	(2)	(3)	(4)	(5)	(6)
	Labor productivity		TFP		TFP	
			Invention patents		Dropping two outliers	
AI treatment (1/0)		0.0796*** (0.0241)		0.1047*** (0.0345)		0.0790*** (0.0324)
AI number (ln(AI+1))	0.0389** (0.0175)		0.0391 (0.0253)		0.0563** (0.0240)	
Other patent treatment (1/0)		0.0545*** (0.0170)		0.0776*** (0.0219)		0.0509** (0.0226)
Other patent number (ln(patent+1))	0.0632*** (0.0082)		0.0674*** (0.0115)		0.0444*** (0.110)	
Firm characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.347	0.343	0.213	0.212	0.211	0.210
Observations	9,607	9,607	10,582	10,582	10,548	10,548

Note: In Columns (1) and (2), we include an incremental covariate: logarithm of fixed capital. Figures in parentheses are robust standard errors. *** $p < 0.01$
** $p < 0.05$.

Table 5
AI Technology and Productivity – Endogeneity Issues

	(1)	(2)	(3)	(4)	(5)	(6)
	IV-2SLS OLS	GMM-DIF	GMM-DIF	GMM-SYS	GMM-SYS	GMM-SYS CL
Productivity (−1)	0.6346*** (0.0082)	0.6467*** (0.0220)	0.4739*** (0.0171)	0.7684*** (0.0262)	0.6473*** (0.0412)	0.5214*** (0.0195)
Productivity (−2)			0.2741*** (0.0164)		0.0036 (0.0121)	
AI number (ln(AI+1))	0.0368* (0.0194)	0.0701*** (0.0244)	0.0413* (0.0214)	0.0050 (0.0353)	0.0579* (0.0312)	0.0571* (0.0315)
Other patent number (ln patent+1))	0.0145 (0.0090)	0.0333*** (0.0094)	0.0262*** (0.0082)	0.0306* (0.0168)	0.0479** (0.0195)	0.0346** (0.0174)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.507					
Number of Instruments		24	24	1041	1137	128
Hansen-Sargan (p-value)	0.367	0.428	0.430	1.000	1.000	0.638
AR(1)		[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
AR(2)		[0.031]	[0.001]	[0.024]	[0.161]	[0.019]
AR(3)			[0.386]			
Observations	9,808	9,812	9,109	9,782	9,714	9,878

Note: Figures in parentheses are standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The values reported for AR(1), AR(2), and AR(3) are the p -values for first, second, and third-order auto-correlated disturbances in the first differences equations. The instrument variable used in Column (1) is R&D expenditure. CL denotes the method of collapsing the instrument matrix developed in Roodman (2009b).

support the productivity-enhancing effect of AI technology. However, as shown in the Hansen-Sargan test, which strongly rejects the null hypothesis with a p -value approaching 1, there is an over-fitting (too many instrument) issue, as argued in Roodman (2009a).¹⁰ Following Roodman (2009b), we collapse the instrument matrix and report the estimates in Column 6.¹¹ The estimated coefficient of AI patents becomes significantly positive at the 10% level, supporting the productivity-enhancing effect of AI technology.

In summary, our robustness checks tend to confirm a considerable productivity effect of AI technology at the firm level, which is consistent with the findings of existing studies such as Alderucci et al. (2020), Ballestar et al. (2020), and Gal et al. (2019). Extant studies seem to support the optimistic view of the productivity effect brought about by

the introduction of AI without facing the so-called modern productivity paradox problem (Brynjolfsson et al., 2019). The possible interpretations are as indicated in Agrawal et al. (2019a): AI may enable firms to find new combinations of existing technologies to promote productivity. The invented AI technologies, such as machine learning, automation, and robots, can also facilitate more productive manufacturing.

5. AI technology and employment

How does AI technology impact the labor market? This crucial issue remains a textual description and conceptual prediction because of the lack of firm-level data (Brynjolfsson et al., 2018). In practice, there may be concerns about the possible labor substitution effect of AI to cause unemployment. Production workers fear that their jobs could be replaced by robots and automatic production. This section further addresses the influence of AI innovations on employment and labor composition within firms as firms invent AI technologies.

¹⁰ A simulation in Windmeijer (2005) demonstrates that the bias of parameters of concerned variables may be significantly reduced by cutting the instrument count in an 8×100 panel. Our dataset comprises 17 periods for approximating 600 firms per year, implying that the number of instruments is important in GMM estimates.

¹¹ The collapse command is subordinate to the xtabond2 command for calculating GMM-SYS estimations using Stata 15.0.

Table 6
AI Technology and Employment

Dep. var.	(1) ln(employment)	(2)	(3)	(4) Employment growth
AI treatment (1/0)		0.0487*** (0.0123)		−0.0005 (0.0219)
AI number (ln(AI+1))	0.0369*** (0.0090)		−0.0057 (0.0160)	
Other patent treatment (1/0)		0.0417*** (0.0086)		0.0200 (0.0154)
Other patent number (ln(patent+1))	0.0367*** (0.0042)		−0.0042 (0.0075)	
Firm sale (ln(sale))	0.2906*** (0.0058)	0.2977*** (0.0058)		
Firm size (ln(employment))			−1.0526*** (0.0124)	−1.0574*** (0.0121)
Firm age	−0.0140*** (0.0012)	−0.0137*** (0.0012)	0.0005 (0.0313)	−0.0023 (0.0313)
Fixed capital	0.3429*** (0.0050)	0.3379*** (0.0047)		
Export intensity	−0.0027*** (0.0002)	−0.0028*** (0.0002)	0.0001 (0.0004)	0.0001 (0.0004)
Number of foreign affiliates	−0.0523*** (0.0152)	−0.0482*** (0.0152)	−0.0473* (0.0273)	−0.0486* (0.0273)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R-squared	0.575	0.571	0.479	0.479
Observations	10,582	10,582	9,596	9,596

Note: Figures in parentheses are robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant term is included.

5.1. The impact of AI on employment and its growth

We first estimate the impact of AI on labor demand and specify a reduced form of employment demand, as follows:

$$\ln \text{Employment}_{it} = \beta_0 + \beta_1 \text{AI}_{it} + \gamma X_{it} + u_t + v_i + \varepsilon_{it} \quad (3)$$

Here, the outcome variable $\ln \text{Employment}$ is the logarithm of the number of employees for firm i in year t . AI is again the AI innovation measured by a treatment measure (0/1) or the number of AI patents. X is a vector of firm characteristics that influence employees' recruitment, mainly non-AI innovations, sales (the logarithm of sales), firm age, fixed capital (the logarithm form), export ratio (export to sale ratio), and number of foreign affiliates. Terms u , v , and ε denote the year fixed effect, firm fixed effect, and error term, respectively.

Although estimating Eq. (3) clarifies the relationship between AI technology and employment level, a more concerning issue is whether AI technology facilitates or hampers employment growth. Thus, we specify the following equation:

$$\Delta \text{Employment}_{it} = \beta_0 + \beta_1 \text{AI}_{i,t-1} + \gamma X_{i,t-1} + u_t + v_i + \varepsilon_{it} \quad (4)$$

where $\Delta \text{Employment}_{it}$ is the employment growth rate measured by the difference in $\ln \text{Employment}$ between two consecutive years, t and $t-1$. AI technology and firm characteristics (X) enter the equation in the form of lagged one year. Referring to firm growth literature, for example, Coad (2009), firm characteristics include firm size (the logarithm of employment), firm age, export ratio (export to sale ratio), and number of foreign affiliates. Non-AI innovations are also included.

Columns 1 and 2 of Table 6 show that AI technology has a positive influence on employment, rather than reducing or not affecting employment levels. This result echoes the findings of Alderucci et al. (2020) and Bughin (2020). According to the estimated value in Column 1, the impact of AI innovation on employment has moderate economic magnitude. A 10% increase in AI patents is associated with a 0.369% increase in employment within inventing electronics firms. Although the economic magnitude of the effect is slightly larger than that of non-AI

Table 7
AI Technology and Workforce Composition

Dep. var.	(1) ln(non-high-skilled workers ratio)	(2)	(3) ln(less-skilled labor ratio)	(4) ln(college-educated labor ratio)
AI treatment (1/0)	−0.0117*** (0.0029)			
AI number (ln(AI+1))		−0.0108*** (0.0021)	0.0214** (0.0098)	−0.0172 (0.0129)
Other patent treatment (1/0)	−0.0018 (0.0020)			
Other patent number (ln(patent+1))		−0.0065*** (0.0010)	−0.0167*** (0.0046)	−0.0422*** (0.0058)
Wage (ln(wage per employee))	−0.0021 (0.0016)	0.0012 (0.0017)	0.1118*** (0.0076)	0.3453*** (0.0200)
Firm Age	−0.0028*** (0.0003)	−0.0026*** (0.0003)	0.0056*** (0.0013)	0.0249*** (0.0017)
Capital intensity	−0.0032** (0.0014)	−0.0028* (0.0015)	−0.0127* (0.0067)	−0.0328*** (0.0090)
Export intensity	−0.0002*** (5.0-e05)	−0.0002*** (5.0-e05)	−0.0012*** (0.0002)	0.0014*** (0.0003)
Number of foreign affiliates	0.0142*** (0.0036)	0.0155*** (0.0036)	0.0011 (0.0164)	0.3090*** (0.0214)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R-squared	0.173	0.177	0.100	0.147
Observations	10,172	10,172	10,557	9,930

Note: Figures in parentheses are robust standard errors. *** $p < 0.01$, ** $p < 0.05$.

Table 8
AI Technology and Employment –Endogeneity Issues

Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)
	GMM-DIF	ln(employment) GMM-SYS	GMM-SYS	GMM-SYS	ln(non-high-skilled workers ratio) GMM-SYS CL	GMM-SYSCL
Lag1 of the dependent variable	0.3028*** (0.0263)	0.8366*** (0.0271)	0.7428*** (0.0289)	0.7430*** (0.0400)	0.5507*** (0.0761)	0.5883*** (0.0792)
Lag2 of the dependent variable						0.0553*** (0.0256)
AI number (ln(AI+1))	0.0621*** (0.0190)	0.0233* (0.0143)	0.0351*** (0.0136)	−0.0193*** (0.0044)	−0.0087* (0.0047)	−0.0021 (0.0026)
Other patent number (ln(patent+1))	0.0777*** (0.0096)	0.0659*** (0.0096)	0.0239** (0.0083)	−0.0065*** (0.0016)	−0.0065*** (0.0023)	−0.0052** (0.0022)
Firm Controls	Yes	No	Yes	Yes	Yes	Yes
Instruments	25	620	1,057	1,208	264	275
Hansen-Sargan (p-value)	0.000	0.033	1.000	1.000	0.070	0.108
AR(1)	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
AR(2)	[0.589]	[0.569]	[0.551]	[0.005]	[0.005]	[0.245]
Observations	9,552	9,830	9,782	9,455	9,543	8,816

Note: Figures in parentheses are robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Firm controls in Columns 1–3 are as those in Tables 6. Because firm sale, one-year lagged firm size, and fixed capital are highly correlated, firm sale is excluded. Firm controls in Column 4–6 are the same as those in Table 7. CL denotes the method of collapsing the instrument matrix developed in Roodman (2009b).

patents, the difference is not statistically significant. The situation is similar when we compare the estimated coefficients of the AI dummy in Column 2.

Why does AI introduction induce greater employment opportunities rather than generate a disemployment effect as feared by the public? From the perspective of AI-inventing firms, patented AI technologies are embedded in key parts and components to produce new products. The increasing demand for these parts and components induces AI patents to have an employment-enhancing effect. Alternatively, if patented AI technologies such as robots and autonomous production are mainly applied to the manufacturing process, they will have an employment-substitution effect. As Taiwan's electronics firms are more effective at developing intermediate components rather than branded final products, the employment-enhancing effect may dominate the employment-substitution effect, resulting in a positive relationship between AI patents and employment. In other words, AI technology has a product-extension effect for generating more jobs in Taiwan's electronics industry. Notably, in addition to AI technology, other innovations also positively affect jobs in Taiwan.

Regarding other covariates, firms with larger sales or those using more fixed capital are associated with more employment, whereas older firms tend to hire fewer workers, *ceteris paribus*. One notable finding is the impact of globalization on employment. Despite the small estimation value, we find that export intensity has a negative impact on employment; that is, the presence of more export-oriented firms is associated with fewer employment. Taiwanese electronics firms have reallocated labor-intensive production lines to China through export-platform FDI since 2002. The development suggests that domestic electronics exporters may shift to exporting technology-intensive products whose production patterns are more capital- and technology-intensive. Thus, export intensity has a negative relationship with employment. For instance, the Foxconn group established many factories and hired approximately one million workers in China to conduct processing exports in 2016. Correspondingly, its parent enterprises hired only approximately 7600 employees and engaged in extremely high exports, with export ratios of 99.6%. As mentioned previously, electronics firms began to conduct outward FDI by reallocating labor-intensive production to developing countries; firms with more foreign affiliates hire fewer employees in parent firms.

However, AI technology and other innovations do not seem to facilitate higher growth rates, as shown in Columns 3 and 4. This result suggests that firms without patents experience similar growth rates through different stimuli. We also find that small firms grow faster than

their larger counterparts, contradicting Gibrat's law (growth and size are independent), while younger firms do not experience a higher employment growth rate. Moreover, establishing more foreign affiliates results in a lower growth rate in parent firms, which is consistent with the findings of Huang et al. (2017).

5.2. The impact of AI on workforce composition

Although AI technology enhances employment within inventing firms, it may change the related mix of skills, which reduces the usage of less-skilled production workers, but facilitates the demand of skilled laborers. (Bughin, 2020; Felten et al., 2019). We further utilize the education level to measure skilled labor to evaluate the skill bias in labor demand caused by AI technology.¹² This issue particularly matters to Taiwan because the higher education expansion policy implemented since the 1990s largely increased the number of university-educated laborers (Chan and Lin, 2015).¹³

Taiwan's key manufacturing industry, the electronics industry, is highly technologically dynamic and recruits a large proportion of young and highly educated workers. We define workers who are graduate-educated as "high-skilled labor", whereas college-and-below educated workers are classified into "non-high-skilled labor". Another reason is that the sharp increase in the supply of college-educated workers casts doubt about the quality of college-educated labor. Moreover, we adopt the conventional description of less-skilled labor as below-college-educated workers (college-and-above educated workers are defined as skilled labor). Table 7 presents the estimation results.

AI technology alters inventing firms' own workforce composition toward lowering the demand for non-high-skilled labor, as the estimated coefficients of AI variables are negative and significant in Columns 1 and 2. The estimated coefficient of the AI dummy in Column 1 suggests that AI-inventing firms hire a lower percentage of workers for non-high-skilled labor (1.17%), relative to firms without AI patents. The development of AI technology relies on high-quality workforces, such as R&D

¹² Due to the availability of data, we cannot test the routine-biased technological change (RBTC) indicated by Goos et al. (2014).

¹³ For example, the number of colleges and universities was 67 in 1996; the number then increased sharply to 149 in 2009 and remained stable thereafter. Correspondingly, the admission rate of joint university entrance examination increased from 49.24% in 1996 to 97.14% in 2009. Although this admission rate declined slightly from 2009 onward, this development suggests that the ratio of college-educated workers in the labor market increased substantially.

personnel and technical engineers; thus, AI technology inventing firms hire more high-skilled labor.

Other patents are developed to lower production costs or promote product quality, and their impacts on workforce composition are theoretically uncertain. Column 2 shows that non-AI patents negatively impact the demand for non-high-skilled labor; a 10% increase in the number of non-AI patents reduces the demand for non-high-skilled labor by 0.065 percentage points. However, the economic magnitude of this negative impact is significantly lower than that of AI technology patents (0.108 percentage points).

Columns 3 and 4 reveal interesting and crucial findings. Column 3 shows that AI technology patents are positively associated with the demand for less-skilled labor, whereas other patents have a negative impact on firms' own share of less-skilled labor. Combined with the estimates in Column 2, innovating firms may adjust their recruitment strategies in response to the increasing supply of college-educated workers. This conjecture is supported by the estimates in Column 4, which show that AI technology negatively impacts the college-educated labor share. In contrast, other patents have no impact on this measure.

The above results tend to indicate a complex and possibly nonlinear relationship between AI invention (non-AI patents) and workforce composition, leading to polarization, as argued by Felten et al. (2019). Our findings also echo Bughin's (2020) argument that AI technology affects the related mix of skills. Following the advent of AI technology, the demand for high-skilled laborers increased, that for middle-skilled (college-educated) workers decreased, whereas that for less-skilled workers remained relatively stable or increased in Taiwan's electronics industry.

Table 8 summarizes the estimation results by adopting various GMM techniques for dynamic panel data to deal with the endogeneity problem for robustness checks.¹⁴

The lagged values of employment and the non-high-skilled worker ratio are both associated with a significantly positive coefficient, indicating path-dependent employment. As shown in Columns 1–3, the estimated coefficients of AI patents are significantly positive, reconfirming the pro-employment effect brought about by AI technology. The Hansen-Sargan test is not rejected at the 1% level in Column 2 whereas the corresponding *p*-value in Column 3 is extremely high, implying an over-fitting issue, as mentioned in Roodman (2009a). The influence of the AI technology on labor composition, is reflected in the estimates obtained by using GMM-SYS and collapsing the instrument matrix in Columns 4–6, which reaffirm the argument that introducing AI technology reduces the number of jobs for non-high-skilled workers.

6. Conclusions

The breakthroughs in AI technologies in the past decade are expected to significantly impact firms, particularly on productivity and labor demand. These impacts are probably just as, or even more significant than, those of the Industrial Revolution, but they are much harder to predict (Makridakis, 2017). Therefore, understanding the economic effects of AI technology requires firm-level data on AI innovation.

Using the keyword-matching method to parse the text of Taiwanese patent documents, we identified AI-related patent classifications, used patents capturing AI innovations, and match-listed electronics firms to AI patents to construct a firm-level panel dataset for the period 2002–2018. This rich data enabled us to examine the impact of AI technology on firm productivity, labor demand, and workforce composition in Taiwan.

Empirical estimations illustrated that AI technology is significantly and positively associated with firm productivity. This finding is

consistent with that of previous empirical studies. Other patents also generate a productivity-enhancing effect of an economic magnitude similar to that of AI technology. After dealing with measurement and outlier issues, the estimates remain similar. We also adopt various techniques of GMM for the dynamic panel data model to deal with endogeneity and obtain similar results. Firm productivity increases by about 5% if an electronics firm increases the number of granted AI patents by 10%.

We also find a pro-employment effect of AI technology and other patents. Notably, AI-inventing firms demand about 3.5% more employees after controlling for non-AI patents. From the perspective of AI-technology-inventing firms, this result is not counterintuitive. In Taiwan's electronics firms, patented AI technologies are mainly used to produce intermediate components and/or final products, rather than improving manufacturing processes. However, AI technology alters firms' workforce composition by favoring more high-skilled (graduate-educated) labor. The influence of AI technology on workforce composition tends to be nonlinear, which is particularly disadvantageous for college-educated workers because they are neither traditional blue-collar production workers nor high-quality R&D personnel or engineers.

Given the rapid development of AI technology and its application in a wide range of production processes and products, our analyses may yield useful implications for R&D and labor policies. In response to the rapid progress of AI technology, the supply of required high-skilled labor seems to be insufficient and lagged. Universities and innovating firms can establish closer ties in cooperative R&D for the development of AI technologies. Many governments have adopted various measures to support the development of specific technologies. Although this is an acceptable strategy and is probably effective in the short run, a uniformly applicable measure such as R&D tax credit is more feasible because other innovations also exhibit similar influences on productivity, employment, and workforce composition, as found in this study.

CRedit authorship contribution statement

This study is a single-authored paper.

Declaration of Competing Interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.respol.2022.104536](https://doi.org/10.1016/j.respol.2022.104536).

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¹⁴ Furthermore, we implemented robustness checks by using the PSM approach for a like-for-like comparison. The estimation results are reported in Table A4 in the Appendix.

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